Violated assumption	Degradation source	Sec. IV Localization	Sec. V Planning	Sec. VI Semantic	Sec. VII Multi-robot	Sec. VIII Faults	Response families
A1. Measurements are informative and correctly associated	Perceptual degradation	38/34/9 W18 R3 N17	3/1/1 W3	2/2/0 W2	4/3/0 W2 G1 N1	6/5/0 W4 R1 N1	IV-A selective updates IV-B complementary sensing IV-E relocalization VIII-B integrity monitors
	Sensor fault and false association	6/5/3 W2 R1 G1 N2	1/0/0 W1	1/1/1 W1	8/6/3 W5 N3	10/10/6 W7 R3	
A2. Wheels roll on supporting ground with known contact	Terrain and wheel-ground contact	17/13/4 W13 R1 N3	20/20/2 W17 R1 G2	4/3/0 W4	3/2/0 W2 G1	5/5/1 W4 N1	IV-C contact priors V-A terrain risk V-E tracking, delay
A3. The mapped scene is static or persistent	Moving agents	16/13/0 W6 R2 N8	23/21/7 W21 G1 N1	8/8/1 W7 G1	–	8/7/0 W6 N2	IV-D persistent maps V-C prediction V-D risk constraints VI-A semantic memory
	Map and scene change	16/16/3 W7 R2 N7	2/2/1 W2	7/6/4 W7	1/1/0 W1	4/0/0 W4	
A4. Learned and semantic inference is valid	Learned and semantic inference error	4/3/1 W1 R2 N1	14/12/1 W13 G1	7/5/0 W7	–	9/8/0 W8 N1	VI-B learned actions VI-C candidate selection VI-D decision timing VIII-B integrity monitors
A5. Shared information is timely and consistent	Communication loss, delay and bandwidth	2/1/0 W1 R1	3/2/2 W3	–	30/18/11 W20 G3 N7	2/1/1 W2	V-E tracking, delay VII-B distributed estimation VII-C rendezvous, repair
A6. Actuation, computation and energy remain available	Actuator fault and saturation	–	5/3/3 W5	1/1/1 W1	5/3/1 W5	8/8/4 W7 R1	VIII-A tolerant control VIII-C executed recovery VIII-D resource management
	Computation and energy limits	4/2/2 W2 N2	4/0/0 W4	9/0/0 W7 G2	6/1/1 W5 N1	7/3/3 W4 R1 N2	

Strongest setting with a named condition: none; replay, simulation, model, or resource; hardware or recorded data; physical closed loop